REPORT 014: EPOCH 014 — 8OPPERC TIMBRE SNAPSHOT POOL & BEAT-SYNCHRONIZED SCENE MORPHING MANAGER

2026-11-04 01:30:38 | Implementation of src/snapshot_pool.rs and js/snapshot-pool-manager.js with JSONL preset pools, named scene JSON files, rehearsal star ratings, selective voice-lane locks, and beat-synchronized scene morphing

Operator: Antigravity (Google DeepMind) · Follows: report-013

Prompt

```
Epoch 14 of 20: Implement the 8opPerc Timbre Snapshot & Scene Morphing Manager in
↪  `src/snapshot_pool.rs` and `js/`: manage durable 8-voice snapshots in JSONL
↪  preset pools (`nanokontrol2-random-preset-pool.jsonl` and legacy
↪  `midimix-random-preset-pool.jsonl`) plus named timbre-state JSON files. Allow the
↪  performer in the `egui` cockpit (or via `nanoKONTROL2` controls and OSC) to
↪  browse saved 8-voice snapshots, rate/favorite snapshots during rehearsal, lock
↪  specific voice lanes (e.g. keep kick and snare while loading hat/clap/tom timbres
↪  from another snapshot), and smoothly morph continuous density/macro parameters
↪  between two scenes over a configurable number of beats. Add automated Rust and
↪  Node.js tests (with `CARGO_BUILD_JOBS=2`), git commit all changes, and compile
↪  `reports/report-014.html` from `reports/report-014.md` via
↪  `/home/renick/claude/html-report-generator/build-report.sh --pdf` and commit.
```

TL;DR

Implemented the 8opPerc Timbre Snapshot & Scene Morphing Manager in src/snapshot_pool.rs and js/snapshot-pool-manager.js: (1) durable 8-voice snapshot management in JSONL preset pools (nanokontrol2-random-preset-pool.jsonl and legacy midimix-random-preset-pool.jsonl) plus standalone named timbre-state JSON files; (2) rehearsal snapshot browsing, 0-5 star ratings, favorites, and selective per-lane voice locks (e.g. locking kick and snare while recalling hat/clap/tom timbres from another snapshot); and (3) smooth beat-synchronized scene morphing interpolating continuous per-lane density, macro/gain, global density, and rhythmic feel between Scene A and Scene B over a configurable number of musical beats. Verified with 54 passing Rust tests and 11 passing Node.js tests (65 total).

Plain-Language Overview: What Changed and Why It Matters

During a live algorithmic percussion performance or rehearsal with `8opPerc`, the performer frequently discovers a great-sounding combination of 8 drum sounds (FM timbres) and rhythmic densities and captures it on the fly by pressing `Marker Set (CC 60)` on the Korg `nanoKONTROL2` (or `Send All` on an Akai `MIDI Mix`). Over several rehearsals, hundreds of these snapshots accumulate in a line-by-line log file (`nanokontrol2-random-preset-pool.jsonl` or the earlier `midimix-random-preset-pool.jsonl`).

However, simply saving snapshots to a file is only half the workflow. In live performance, the musician needs to:

1. **Browse, Rate, and Star Snapshots (`SnapshotPoolManager`)**: Quickly filter captured snapshots by star rating (`1` to `5`), favorites ( ), or search keywords (`gqom`, `kick-laser`, `minimal`), and save favorite setups as standalone named `.json` scene files.
2. **Mix-and-Match Voices with Per-Lane Locking (`lane_locks`)**: Often a performer loves their

current `kick` and `snare` groove and wants to keep those two lanes anchored while loading the `hat_closed`, `hat_open`, `clap`, `tom`, `rim`, and `cymbal` sounds from a completely different snapshot.

3. **Smoothly Morph Between Two Scenes Over Musical Beats (`SceneMorphEngine`)**: Instead of an abrupt jump from a sparse intro scene (`Scene A`) to a high-energy climax scene (`Scene B`), the performer can select a musical duration (such as `2`, `4`, `8`, `16`, or `32` beats) and let the engine smoothly interpolate every unlocked lane's rhythmic density (`0.1x` to `5.0x`), macro/gain slider (`0..127`), global density, and swing/humanize feel scale across those beats.

In Epoch 14, we implemented this complete system in both Rust (`src/snapshot_pool.rs`) and Node.js (`js/snapshot-pool-manager.js`).

1. Architecture & Plain-Language Mapping Table

The table below connects each plain-English capability to its exact Rust (`src/snapshot_pool.rs`) and Node.js (`js/snapshot-pool-manager.js`) implementation components:

Capability	Plain-English Explanation	Rust Component (`src/snapshot_pool.rs`)	Node.js Component (`js/snapshot-pool-manager.js`)
JSONL Preset Pool Compatibility	Reads and writes both `nanokontrol2-random-preset-pool.jsonl` (`kick..cymbal` keys) and legacy `midimix-random-preset-pool.jsonl` (`kicks, snares, hats, toms, bells, metals, noise, fx` keys)	`SnapshotSceneRecord::from_json_value`, `import_jsonl_str`, `export_jsonl_str`, `append_snapshot_to_jsonl_file`	`normalizeSnapshotRecord`, `formatSnapshotJsonlLine`, `importJsonlString`, `loadJsonlFile`, `appendSnapshotToJsonlFile`
Named Timbre-State JSON Files	Saves and loads standalone, human-readable `.json` scene files containing all 8 timbres, densities, gains, mutes, solos, global feel, ratings, and tags	`save_named_scene_json_file`, `load_named_scene_json_file`, `to_pretty_json`	`saveNamedSceneJsonFile`, `loadNamedSceneJsonFile`
Rehearsal Ratings & Favorites	Assigns `0..=5` star ratings and favorite flags to snapshots and filters the browser by minimum rating, favorite status, or text search	`set_rating`, `toggle_favorite`, `filtered_snapshots`	`setRating`, `toggleFavorite`, `getSnapshot`
Selective Voice-Lane Locking	Locks any subset of the 8 percussion lanes (e.g., locking `kick` and `snare`) so recalling a snapshot or morphing scenes only updates unlocked lanes	`SnapshotPoolManager::lane_locks`, `set_lane_lock`, `recall_snapshot_into_model`	`SnapshotPoolManager.laneLocks`, `setLaneLock`, `recallSnapshotIntoModel`

Capability	Plain-English Explanation	Rust Component (‘src/snapshot_pool.rs‘)	Node.js Component (‘js/snapshot-pool-manager.js‘)
Beat-Synchronized Scene Morpher	Interpolates continuous densities, macro gains, global density, and feel scale from Scene A to Scene B over configurable beats (`2..32` beats) or manual crossfader	`SceneMorphEngine`, `MorphedSceneFrame`, `start_morph`, `advance_beats`, `apply_current_morph_to_model`	`startMorph`, `advanceMorphBeats`, `evaluateMorphFrame`
Interactive `egui` Cockpit Panel	Renders the 8-voice lock bar, filterable snapshot list with star rating buttons, and Scene A/B beat morphing deck	`render_snapshot_pool_panel`	Synced via UDP OSC `/nkeditor/op8/snapshot/save` & `/nkeditor/op8/*`

2. Legacy ‘midimix-random-preset-pool.jsonl‘ & ‘nanokontrol2-random-preset-pool.jsonl‘ Normalization

Snapshots captured in earlier `8operc-konduktiva` sessions (`midimix-random-preset-pool.jsonl`) store their `timbres` map using `KONDUKTIVA_FAMILY_ORDER` keys (`kicks`, `snares`, `hats`, `toms`, `bells`, `metals`, `noise`, `fx`) or live player labels (`percussion`, `alt-toms`), whereas `nanokontrol2-random-preset-pool.j` uses canonical `FAMILY_ORDER` keys (`kick`, `snare`, `hat_closed`, `hat_open`, `clap`, `tom`, `rim`, `cymbal`) or 8-element arrays.

Both `parse_timbres_field` in `src/snapshot_pool.rs` and `normalizeSnapshotRecord` in `js/snapshot-pool-manager.js` route every key through `resolve_lane_index` (`0..8`), guaranteeing that snapshots from either controller pool load into the exact same 8 voice lanes without data loss and hash to the exact same deterministic FNV-1a `pool_bank_index` (`0..4096`).

Tests

QA

```
**Total Tests**: `65` passing (`54` Rust tests + `11` Node.js tests), `0` failing, `0` compiler warnings.

**Safety & Resource Discipline**: All builds and tests executed with `CARGO_BUILD_JOBS=2` in headless/dry-run mode without opening blocking GUI windows or starting live audio playback.

**Verdict**: **PASS**.
```

Files touched

- ‘src/snapshot_pool.rs‘ (Created — 784 lines): Implemented ‘SnapshotSceneRecord‘, ‘SceneMorphEngine‘, ‘MorphedSceneFrame‘, ‘SnapshotPoolManager‘, ‘render_snapshot_pool_panel‘, and 4 unit and headless ‘egui‘ tests.
- ‘js/snapshot-pool-manager.js‘ (Created — 385 lines): Implemented ‘normalizeSnapshotRecord‘, ‘formatSnapshotJsonlLine‘, ‘timbreSetSignature‘, and ‘SnapshotPoolManager‘ with JSONL/JSON file I/O, selective lane locking, and beat-synchronized scene morphing.
- ‘src/lib.rs‘ (Modified): Exported ‘pub mod snapshot_pool;‘.
- ‘js/index.js‘ (Modified): Re-exported ‘./snapshot-pool-manager.js‘.

- ‘js/nkeditor-sdk.test.js‘ (Modified): Added automated end-to-end Node.js test verifying ‘nanokontrol2-random-preset-pool.jsonl‘, legacy ‘midimix-random-preset-pool.jsonl‘, named scene ‘.json‘ persistence, rehearsal ratings/favorites, selective ‘kick‘ + ‘snare‘ lane locking, and 8-beat scene morphing.
- ‘reports/report-014.md‘, ‘reports/report-014.html‘, ‘reports/report-014.pdf‘ (Created): Epoch 014 report in Markdown, grayscale HTML, and PDF formats.

Test coverage (65 total)

QA verdict

PASS

Report 014 | Generated 2026-11-04 01:30:38 | reports/report-014.html